

# Rakan Bani-Melhem

## Data Scientist

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## Professional Summary

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Master's-educated Data Scientist with 4 years of expertise in statistical models, machine learning, and data visualization. Skilled in Python, tuning supervised and unsupervised machine learning models, and building dashboards with analytical storytelling. Proven track record of deploying automated high-performance machine learning pipelines and scalable dashboards that improve business performance through data-driven insights. Passionate about transforming complex data into actionable insights that drive smarter decision-making.

## Experience & Key Achievements

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### ScoreFex Intelligent Football Analysis | Full-Time | Jul 2025 - Jul 2026

#### *Data Scientist*

- Delivered a strong ROI within 4 months by implementing machine learning workflows that automated predictive models for football match outcomes.
- Increased revenue through enhanced customer engagement by developing and deploying real-time football analysis dashboards that enabled customers to extract key insights effortlessly.
- Substantially reduced financial risk by leading the development of strategic sports odds analytics solutions.

### Accuyield SDN BHD

#### *Machine Learning Engineer* | Full-Time | Jan 2025 - Jul 2025

- Increased ROI by optimizing and deploying existing machine learning models through statistical analysis and model selection strategies.
- Significantly reduced machine learning processing time through implementation of data science best practices and optimized feature engineering techniques.
- Improved operational efficiency by automating machine learning workflows using Airflow and Docker, deploying predictive models to cloud platforms with monitoring systems that ensured reliability and stability.

#### *Data Scientist* | Full-Time | Mar 2022 - Dec 2024

- Automated ETL data pipelines to significantly reduce costs, increased data accuracy by 9%, while substantially cutting data processing time.
- Deployed end-to-end machine learning pipelines across multiple domains to support decision-making and drive meaningful business growth in return on investment.
- Developed analytical dashboards and reports to analyze data and monitor model performance that highlighted critical findings and trends, supporting data-driven strategic decisions.

## Education

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### MSc in Data Science & Business Analytics | Full-Time | Jan 2021 - Jan 2022

- Dual Degree awarded jointly by Asia Pacific University and De Montfort University, England.

Specializing in Data Science and Machine Learning, with a strong foundation in ETL Pipeline Management, MLOps, and Data Analytics to drive intelligent data-driven solutions and predictive modeling across diverse domains.

## Technical & Soft Skills

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- **Technical Skills:** scikit-learn, TensorFlow, PyTorch, AWS, Airflow, SQL, AI Agents, Generative AI, NLP.
- **Analysis Skills:** Analysis and Storytelling, Trend Analysis, Analytical Reports, EDA, Tableau, Power BI, Excel.
- **Soft Skills:** Data Driven, Decision Making, Problem Solving, Team Collaboration, Adaptability, Accountability.
- **AI-Assisted Development:** Claude Code, Claude Cowork, Codex, Anthropic API, OpenAI API, MCP.

## Projects Experience

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### Data Integration and ETL Process Management

- Built and managed ETL pipelines that integrated data from multiple sources including APIs and web platforms.
- Implemented automated processes to clean and transform data to ensure quality data for reporting and analytics.

### Predictive Model Development

- Delivered accurate classification and regression predictions by training and validating supervised machine learning.
- Developed unsupervised machine learning using factor analysis and cluster analysis to reduce data complexity.

### Automated Workflows and Model Deployment

- Deployed automated ETL and ML workflows on cloud platforms to ensure scalability and reproducibility.
- Established monitoring systems to track model performance, detect data drift, and trigger retraining workflow.

### Data Visualization and Report Analytics

- Developed and deployed user-friendly dashboards that allowed customers to extract key insights quickly.
- Implemented analytical dashboards that helped stakeholders analyze data and monitor model performance.
- Delivered analytical reports that highlighted critical data analysis findings, model evaluation, and trends.

### Forecasting and Trend Analysis

- Developed and deployed statistical forecasting models using regression and machine learning to support marketing strategy by predicting revenue, performance, and demand trends.
- Built time series forecasting models using LSTM neural networks to forecast financial market trends and patterns.

### NLP Analytics and Insights Extraction

- Applied NLP techniques including sentiment analysis and text classification to extract key insights by analyzing reviews and social media data.
- Built an AI chatbot using Retrieval-Augmented Generation (RAG) and GPT models to automate customer support and improve response efficiency.

### LLM Applications and AI Agent Development

- Built AI agent workflows using LLM APIs and MCP tooling to automate multi-step data retrieval, analysis, and report generation.
- Developed Retrieval-Augmented Generation (RAG) pipelines over domain-specific document sources to deliver accurate, context-aware responses.